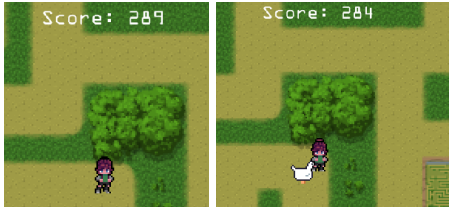


Test Tables

Below, we specify in more detail the setup, expected outcomes and actual outcomes of the manual tests carried out. Where relevant, we also provide evidence, such as screenshots of expected events or visuals. The tests are highlighted in **green** on success, **yellow** on partial success, and **red** on failure.

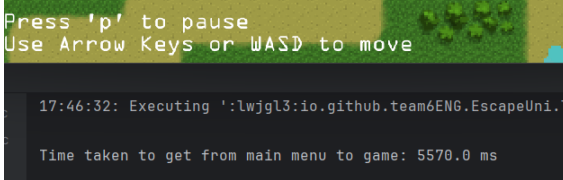
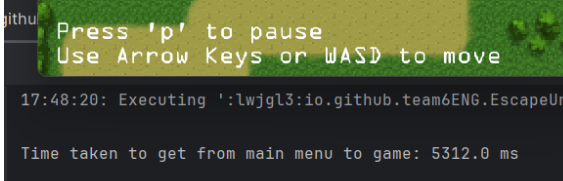
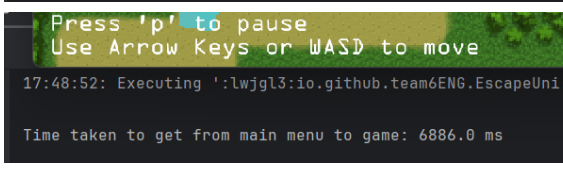
Following the tests is a table of definitions regarding actions taken by the tester whilst playing the game or events that occur in the game: these were designed so that test descriptions can be made more brief and readable without losing clarity. These will be written in *italics* where referenced below.

MT_Collision	
Test Description	The game is <i>arbitrarily set up</i> . The player <i>walks</i> upwards for 10 seconds. Then, they <i>walk</i> right for 5 seconds.
Expected Result	The player should move straight upwards to the edge of the path available and then should stop moving, even whilst the W key is pressed. Similarly, the player should not move beyond the edge of the path on the right, even whilst the D key is pressed.
Actual Result	The player moved straight upwards to the edge of the path available and then stopped moving. The player also could not move beyond the boundary on the right of the path.
Tester(s)	James Ingram
Operating System	Windows 11
Date	19/12/2025
Evidence	

MT_Resize	
Test Description	The game is <i>arbitrarily set up</i> . The user resizes their window to be larger. Then, the user resizes their window to be smaller. Finally, the user stretches their window to be wider but not taller.

Expected Result	The game's graphics should remain in proportion with the size of the window (that is, the shortest of its width and height) at all times, but the graphics should not appear stretched or squashed.
Actual Result	The game's graphics scale appropriately for all of the tested window sizes without appearing stretched or squashed.
Tester(s)	James Ingram
Operating System	Windows 11
Date	19/12/2025
Evidence	

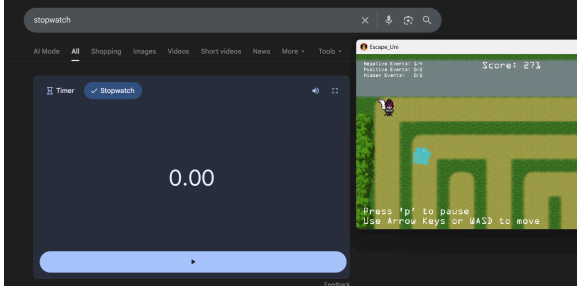
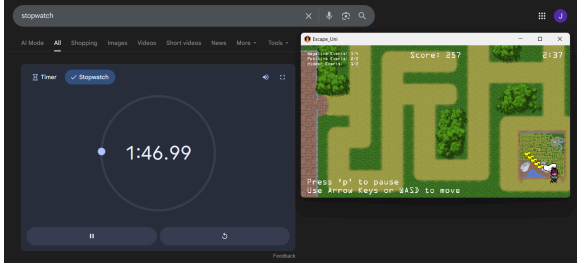
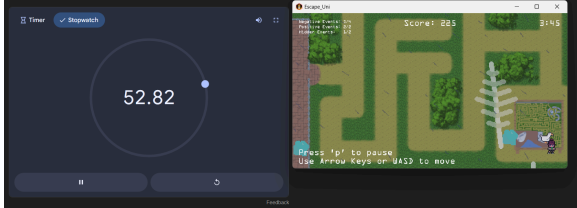
MT_Start_Time	
Test Description	The game executable is run. The play button is pressed. "Testing" is typed into the username box, and the confirm button is pressed. One of the two player characters is arbitrarily selected. Play is pressed on the instructions screen.
Expected Result	The elapsed time from the main menu opening to the game screen showing, printed in the console, should be less than 8 seconds (8000 ms).

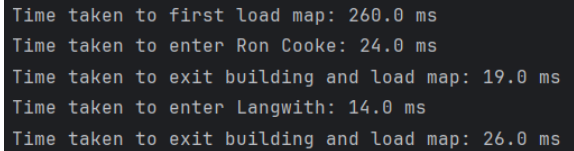
Actual Result	The printed times elapsed for three separate tests were 5570ms, 5312ms and 6886ms respectively on a standard laptop.
Tester(s)	James Ingram
Operating System	Windows 11
Date	04/01/2026
Evidence	  

MT_Pause_Time	
Test Description	The game is arbitrarily set up. The P key is pressed on the keyboard such that the pause menu is opened onto the screen.
Expected Result	The elapsed time between the keypress being detected and the pause menu opening should be less than one second (1000ms).
Actual Result	The printed times for 10 separate tests were all less than 15ms.
Tester(s)	James Ingram
Operating System	Windows 11
Date	04/01/2026

Evidence	<pre> Time taken to get to pause menu after pressing P: 14.0 ms Time taken to get to pause menu after pressing P: 8.0 ms Time taken to get to pause menu after pressing P: 9.0 ms Time taken to get to pause menu after pressing P: 7.0 ms Time taken to get to pause menu after pressing P: 8.0 ms Time taken to get to pause menu after pressing P: 6.0 ms Time taken to get to pause menu after pressing P: 9.0 ms Time taken to get to pause menu after pressing P: 7.0 ms Time taken to get to pause menu after pressing P: 7.0 ms Time taken to get to pause menu after pressing P: 7.0 ms </pre>
-----------------	---

MT_Frame_Rate	
Test Description	The user <i>wins</i> the game and regularly checks their recent average frame rate. After winning, the player reads their average frame rate across the whole game.
Expected Result	The average frame rate should be at least 30 frames per second at all times, and at least 60 frames per second across 99% of readings.
Actual Result	The frame rate did not descend below 75 frames per second, even whilst other applications were open on the device.
Tester(s)	James Ingram
Operating System	Windows 11
Date	05/01/2026
Evidence	<pre> Average FPS (Recent): 146 Average FPS (Recent): 145 Average FPS (Recent): 154 Average FPS (Recent): 146 Average FPS (Recent): 176 Average FPS (Recent): 93 Average FPS (Recent): 81 Average FPS (Recent): 100 Average FPS (Recent): 77 Average FPS (Recent): 79 Average FPS (Recent): 79 Average FPS (Recent): 80 Average FPS (Recent): 79 Average FPS (Recent): 79 Average FPS (Recent): 80 Average FPS (Recent): 79 Average FPS (Recent): 79 Average FPS (Recent): 80 Average FPS (Recent): 80 Average FPS (Recent): 77 </pre> <pre> Average FPS: 115.05732 Score submitted to remote server. Position: 28 Retrieved scores from remote server </pre>

MT_Navigation	
Test Description	The user <i>arbitrarily sets up</i> the game and <i>walks</i> to the top left corner of the map. They start a stopwatch, and attempt to <i>walk</i> to the bottom right corner of the map, stopping the timer once they arrive.
Expected Result	Over three attempts taking different routes, the player stopwatch should never exceed 2 minutes and 30 seconds.
Actual Result	Attempt 1: 1 minute and 47 seconds Attempt 2: 1 minute and 7 seconds Attempt 3: 53 seconds
Tester(s)	James Ingram
Operating System	Windows 11
Date	05/01/2026
Evidence	    The evidence consists of four screenshots from a game titled 'Escape Lyn'. Each screenshot shows a stopwatch on the left and a game map on the right. The stopwatch times are 0.00, 1:46.99, 1:06.32, and 52.82 respectively. The game map shows a character moving through a maze-like environment. The score is 273, 257, 203, and 205 respectively. The instructions 'Press 'p' to pause' and 'Use Arrow Keys or WASD to move' are visible on the map.

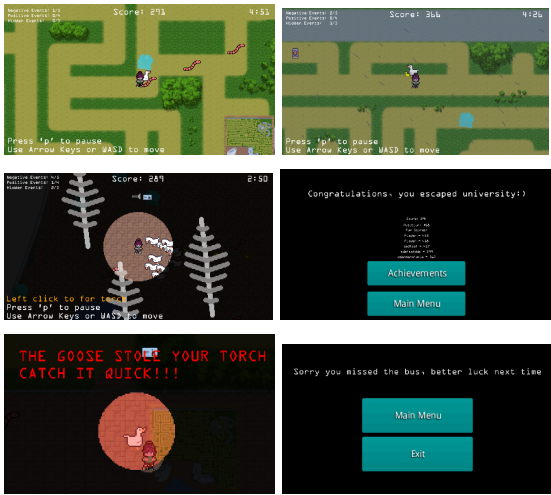
MT_Load	
Test Description	The user <i>arbitrarily sets up</i> the game and <i>wins</i> . The terminal outputs times taken for the map to initially load, as well as the times to <i>enter</i> and <i>exit</i> buildings, throughout this.
Expected Result	All of the times taken to enter and exit buildings measured, as well as the time taken to load the map initially, should be at most 1 second (1000 ms).
Actual Result	All times were no more than half a second.
Tester(s)	James Ingram
Operating System	Windows 11
Date	06/01/2026
Evidence	 <pre> Time taken to first load map: 260.0 ms Time taken to enter Ron Cooke: 24.0 ms Time taken to exit building and load map: 19.0 ms Time taken to enter Langwith: 14.0 ms Time taken to exit building and load map: 26.0 ms </pre>

MT_Min_Time	
Test Description	The user <i>arbitrarily sets up</i> the game and <i>wins</i> three times. After each win, the user records the time taken to win given by the game.
Expected Result	None of the win times are 2 minutes or less.
Actual Result	First time: 3 minutes, 13 seconds Second time: 3 minutes, 39 seconds Third time: 2 minutes, 16 seconds
Tester(s)	James Ingram
Operating System	Windows 11
Date	07/01/2026

Evidence

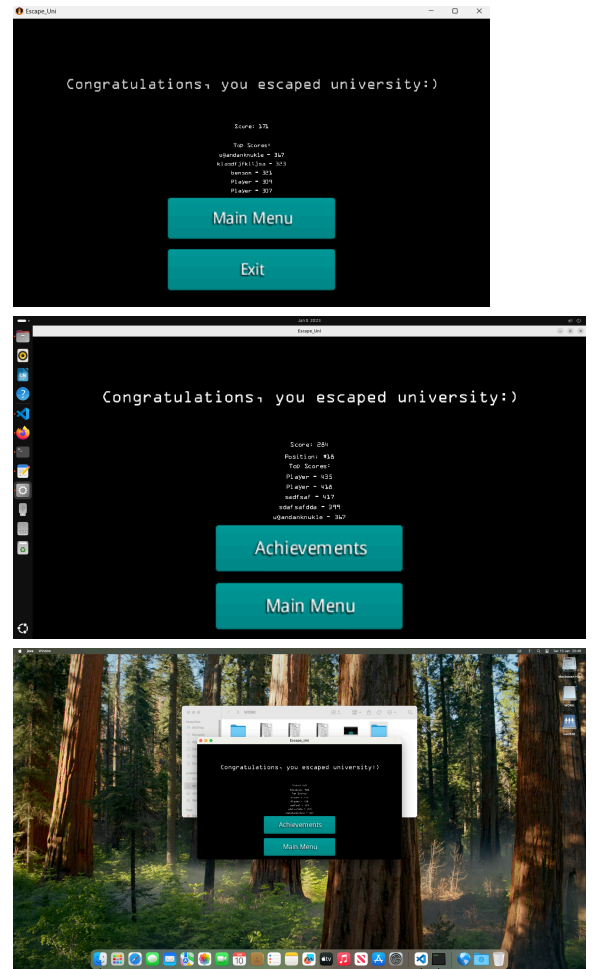


MT_No_Crash	
Test Description	The user <i>arbitrarily sets up</i> the game and either <i>wins</i> or <i>loses</i> 100 times. The game may be altered so that the player's speed is increased for up to 80 of these tests. The player must lose for at least 10 tests. All events occur on at least 3 playthroughs.
Expected Result	On none of the playthroughs should the game crash, that is, unexpectedly close due to an error.
Actual Result	The outcomes of the games were distributed as follows: - <i>Won</i> the game at usual speed. (10%) - <i>Lost</i> the game (10%) - <i>Won</i> the game with increased player speed (80%). All events occurred on at least 3 playthroughs. The game did not crash on any playthrough.

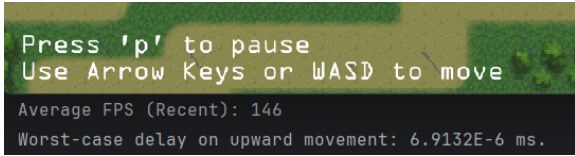
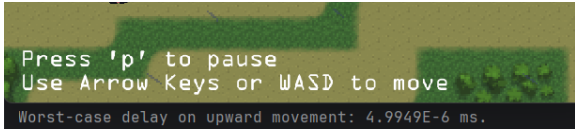
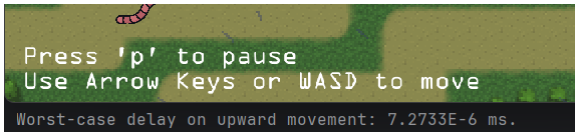
Tester(s)	James Ingram
Operating System	Windows 11
Date	07/01/2026 - 10/01/2026
Evidence	

MT_OS_Compatibility	
Test Description	The game is <i>arbitrarily set up</i> on a PC using the 3 . The user <i>wins</i> the game.
Expected Result	The game should not crash throughout gameplay and should not exhibit any unusual behaviour seemingly unique to this operating system.
Actual Result	The game does not crash throughout gameplay and does not appear to exhibit any unusual behaviour unique to this operating system.
Tester(s)	James Ingram, Olivia Spencer
Operating System	Windows 11, Ubuntu LTS, Mac OS
Date	19/12/2025, 8/1/2026, 10/1/2026

Evidence



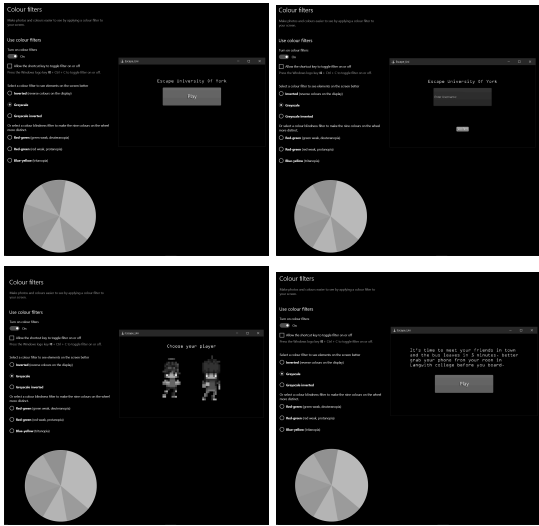
MT_Input_Delay	
Test Description	The game is arbitrarily set up. The W key is pressed to move upwards, and the program provides the <i>worst-case input delay on W</i> .
Expected Result	The worst-case input delay on W initially measured should not exceed 1ms.
Actual Result	The printed times for 5 separate tests were all significantly less than 1ms.
Tester(s)	James Ingram
Operating System	Windows 11
Date	05/01/2026
Evidence	The evidence is a screenshot of a game screen. It shows a green field with a small red object. The text on the screen reads: "Press 'p' to pause", "Use Arrow Keys or WASD to move", and "Worst-case delay on upward movement: 6.4557E-6 ms.".

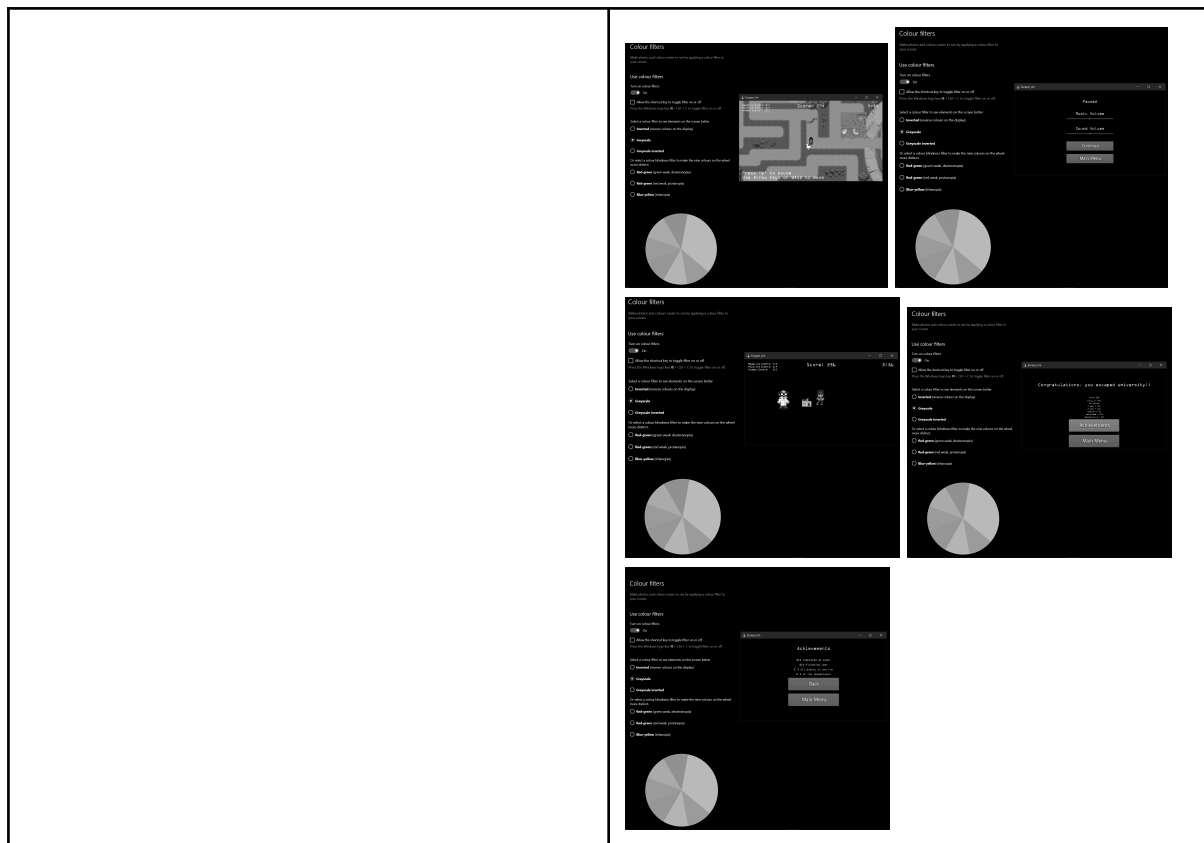
	   
--	---

MT_Memory	
Test Description	The game is arbitrarily set up and the user wins the game, whilst running a task manager on their OS. If the memory usage by the program exceeds 500MB then this is noted by the user.
Expected Result	The user should not notice memory usage exceeding 500MB.
Actual Result	The memory usage never exceeded 300MB.
Tester(s)	James Ingram
Operating System	Windows 11
Date	05/01/2026
Evidence	

MT_Inter-Menu	
Test Description	The game is arbitrarily set up and the user wins the game. In the terminal, the elapsed times to load from one menu to the next are printed.
Expected Result	For loading time between any two menus during gameplay, the elapsed time printed should not be above 16 ms.

Actual Result	The time to load from one menu to another was found to be at most 13 ms throughout gameplay.
Tester(s)	James Ingram
Operating System	Windows 11
Date	10/01/2026
Evidence	Time taken to get from main menu screen to username screen: 12.0 ms Time taken to get from username screen to character selection screen: 13.0 ms Time taken to get from character select screen to instructions screen: 8.0ms Time taken to get from pause screen to main menu screen: 5.0 ms

MT_Accessibility	
Test Description	The game is <i>arbitrarily set up</i> and <i>won</i> with a greyscale filter applied.
Expected Result	The game should have the instructions and text be legible with a greyscale applied. It should hence be winnable with said filter.
Actual Result	The game does have the instructions and text both legible with a greyscale applied. Hence it is winnable with said filter.
Tester(s)	Olivia Spencer
Operating System	Windows 10
Date	11/01/2026
Evidence	 The evidence consists of four screenshots of the game's accessibility settings menu. Each screenshot shows the 'Use colour filter' option set to 'On' and the 'Adjust intensity of filter' slider set to 100%. The screenshots are arranged in a 2x2 grid. The top-left screenshot shows the 'Use colour filter' option set to 'On' and the 'Adjust intensity of filter' slider set to 100%. The top-right screenshot shows the 'Use colour filter' option set to 'On' and the 'Adjust intensity of filter' slider set to 100%. The bottom-left screenshot shows the 'Use colour filter' option set to 'On' and the 'Adjust intensity of filter' slider set to 100%. The bottom-right screenshot shows the 'Use colour filter' option set to 'On' and the 'Adjust intensity of filter' slider set to 100%.



Game Action, Event and Quantity References

Name of Action, Event or Quantity	Description
<i>Arbitrarily set up</i>	The game executable is loaded. The play button is pressed. Some username (which is not of importance) is typed into the box, and the confirm button is pressed. One of the two player characters is arbitrarily selected. Play is pressed on the instructions screen.
<i>Walks</i>	The player uses a combination of the keys W, A, S, D and the arrow keys to control the player with the goal of moving in a given direction or towards a given location.
<i>Wins</i>	The player <i>arbitrarily sets up</i> the game and then takes any sequence of actions leading to the win screen being loaded.
<i>Loses</i>	The player <i>arbitrarily sets up</i> the game and waits until the timer causes them the game over screen being loaded.
<i>Worst-case input delay on W</i>	The worst case for initial input delay is that

	the user first pressed W right after the input was checked for on the previous frame. In this case, the time taken to respond to the keypress would be the time since the previous frame plus the time taken to move the player after the keypress was detected. Hence, this is the formula used to calculate this “worst-case input delay”.
<i>Enters building</i>	The user presses G with sufficiently small distance to the referenced building (i.e. Ron Cooke Hub, Langwith, Piazza) on the map and a new screen is loaded.
<i>Exits building</i>	After having <i>entered</i> a building, the user eventually presses G, causing the original game screen to load in almost the same state as it originally had before <i>entering</i>.